Quantification: 543_3
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	13
interior lacuna count	12
frame-edge lacuna count	1

Interior lacunae only: mean, median and sample SD (n = 12)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2350.2	2188.5	668.925	px^2
major_axis_length_px	lacuna shape	91.5632	84.5141	33.0603	px
minor_axis_length_px	lacuna shape	36.5948	35.0553	10.2044	px
aspect_ratio	lacuna shape	2.6601	2.3867	1.276	unitless
eccentricity	lacuna shape	0.889	0.9075	0.078	unitless
solidity	lacuna shape	0.8742	0.9204	0.1013	unitless
roots_count	headline	4.3333	3	4.1194	count
ring_length_r30_px	headline	243.333	261	146.182	px
ring_length_r60_px	reported	765.917	812	300.452	px
owned_length_px	ownership-dependent	2356.88	1744.02	2164.2	px
edge_count	ownership-dependent	106.333	95.5	95.0352	count
mean_edge_length_px	ownership-dependent	17.4789	21.4958	9.1546	px
perimeter_px	normalised	221.015	219.693	100.917	px
ring_area_r30_px2	normalised	8668.5	8576.5	2568.76	px^2
ring_area_r60_px2	normalised	21815.7	21875	4394.01	px^2
in_frame_fraction_r30	normalised	0.9974	1	0.009	unitless
in_frame_fraction_r60	normalised	0.9801	1	0.058	unitless
ring_density_r30	normalised	0.0257136	0.0286336	0.011897	px^-1
ring_density_r60	normalised	0.0340131	0.0331213	0.00828064	px^-1
roots_per_100px_perimeter	normalised	1.6316	1.5257	1.3339	per 100 px
ring_attached_length_r30_px	network v2	188	152	151.56	px
ring_attached_length_r60_px	network v2	495.75	504.5	358.966	px
ring_length_w_r30_px	network v2	274.857	296.507	167.72	px
ring_length_w_r60_px	network v2	872.163	920.059	340.689	px
sholl_crossings_r10	network v2	5.8333	5	4.3658	count
sholl_crossings_r20	network v2	9.3333	11	4.7162	count
sholl_crossings_r30	network v2	10.4167	10.5	4.814	count
ring_width_mean_r30_px	width	6.9985	7.2994	1.103	px

Per field
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measure	value	units
analysed_area_px2	1.02406e+06	px^2
total_skeleton_length_px	35656	px
canalicular_length_density_per_px	0.0348182	px^-1
junction_count	873	count
junction_density_per_px2	0.00085249	px^-2
lacuna_count	13	count
lacunae_per_px2	1.269e-05	px^-2
skeleton_component_count	167	count

mean_component_length_px	213.509	px
median_component_length_px	18	px
field_length_density_w_per_px	0.0396596	px^-1
field_density_without_flagged_per_px	0.0348182	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	224	px
unattached_component_count	15	count
canalicular_length_density_attached_per_px	0.0348182	px^-1
field_length_density_with_unattached_per_px	0.0350369	px^-1

Canalicular width, the new measure

measure	value	units
width_mean_px	7.7284	px
width_median_px	8	px
width_p10_px	4.4721	px
width_p90_px	10	px
width_pixels_used	35656	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2174	80.8669	35.2412	2.2947	0.9	0.9518	0.4109	79.3965	270.898	False
2	3447	90.3677	63.9486	1.4131	0.7066	0.674	0.4223	197.059	916.902	False
3	1883	69.2617	34.8694	1.9863	0.864	0.9448	0.5742	250.833	340.47	False
4	3536	181.867	30.1414	6.0338	0.9862	0.7148	0.0738	327.046	42.4075	False
5	2052	69.2015	39.4228	1.7554	0.8219	0.9112	0.0622	388.671	369.939	False
6	2266	88.1613	35.5691	2.4786	0.915	0.9021	0.1324	473.05	745.191	False
7	1625	80.1248	27.6985	2.8927	0.9383	0.8325	0.2321	545.572	877.076	False
8	1683	71.0123	31.3735	2.2634	0.8971	0.9355	-0.0518	634.68	198.315	False
9	2203	91.2092	31.4053	2.9043	0.9389	0.9295	0.0832	734.301	635.44	False
10	2633	93.5592	36.2778	2.579	0.9218	0.9454	0.0457	780.576	481.495	False
11	531	28.7881	24.8252	1.1596	0.5063	0.9332	0.5942	1012.28	184.834	True
12								458.12	481.03	False
13								857.425	285.616	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	3	257	859	936.477	44	21.2836	7.0427
2	13	518	1368	4580.38	211	21.7079	7.3395
3	3	168	572	980.352	57	17.1992	7.5801
4	8	393	910	5598.03	252	22.2144	7.7496
5	6	265	616	3942.86	169	23.3305	7.098
6	10	306	916	3163.05	139	22.7557	8.1353
7	1	172	837	2507.69	134	18.7141	6.9851
8	2	107	504	10.8284	1	10.8284	7.2593
9	4	383	1083	5588.89	234	23.8841	7.5455
10	2	265	787	974.009	35	27.8288	7.498
11	0	79	410	0	0	0	6.0343

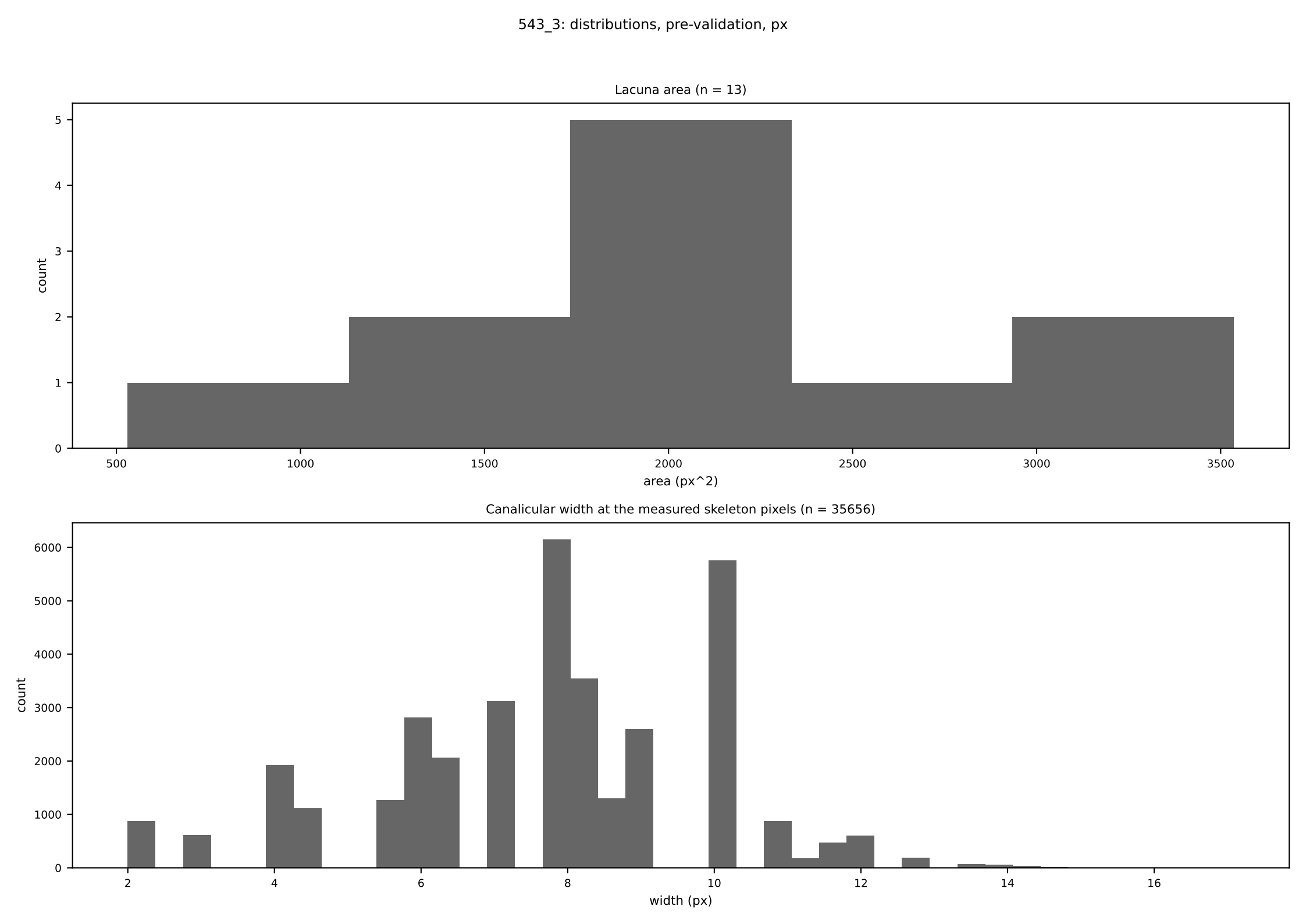
12	0	4	248	0	0	0	4.0322
13	0	82	491	0	0	0	5.7164

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	210.066	8595	21715	1	0.9616	0.0299011	0.0395579	1.4281
2	363.446	12097	29071	1	1	0.0428205	0.0470572	3.5769
3	184.794	7941	19515	1	1	0.021156	0.0293108	1.6234
4	415.99	13536	26753	0.9687	0.7992	0.0290337	0.0340149	1.9231
5	205.965	8333	20013	1	1	0.0318013	0.03078	2.9131
6	241.907	9380	23807	1	1	0.0326226	0.0384761	4.1338
7	229.321	8558	22035	1	1	0.0200981	0.037985	0.4361
8	191.965	7943	21525	1	1	0.013471	0.0234146	1.0419
9	238.108	9212	24077	1	1	0.0415762	0.0449807	1.6799
10	242.936	9386	24420	1	1	0.0282335	0.0322277	0.8233
11	95.3909	3354	9525	0.6102	0.5725	0.023554	0.0430446	0
12	47.4558	4187	13513	1	1	0.00095534	0.0183527	0
13	80.2254	4854	15344	1	1	0.0168933	0.0319995	0

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	110	543	299.149	1006.32	4	12	15
2	469	1167	595.115	1556.15	13	16	19
3	118	386	183.64	647.973	4	7	9
4	382	847	440.848	1034.31	13	13	9
5	238	543	293.865	707.784	7	11	11
6	272	665	341.936	1041.09	7	11	10
7	67	216	192.681	948.181	2	7	11
8	63	154	115.255	561.789	3	4	6
9	333	903	435.948	1220.55	10	14	15
10	186	466	302.764	891.938	6	12	13
11	6	6	84.3553	467.061	1	3	7
12	0	0	3	292.806	0	0	1
13	18	59	94.0833	557.073	1	5	6


543_3: distributions, pre-validation, px
Lacuna area (n = 13)
count
area (px^2)
0
1
2
3
4
5
500
1000
1500
2000
2500
3000
3500
Canalicular width at the measured skeleton pixels (n = 35656)
count
width (px)
0
1000
2000
3000
4000
5000
6000
2
4
6
8
10
12
14
16